

Finite Group Theory

Lectured by Kalyani Kansal

Scribed by Yu Coughlin

Autumn 2025

Contents

1	Introduction	1
1.1	Basic constructions	1
1.2	Actions	2
2	Composition Series	2
3	Finite simple groups	2
4	Sylow's Theorems	2
5	Extensitons	2

1 Introduction

1.1 Basic constructions

A recap of some basic notions we will assume familiarity with going forward.

Proposition 1.1.1. Let $\phi : G \rightarrow H$ be a group homomorphism, then there is a canonical isomorphism:

$$G/\ker(\phi) \cong \text{im}(\phi)$$

Proposition 1.1.2. Let $N \trianglelefteq G$, then every subgroup of G/N is of the form G/H for some $N \leq H \leq G$.

Definition 1.1.3. A subgroup $H \leq G$ is *characteristic* if for all $\phi \in \text{Aut}(G)$, $\phi(H) = H$.

This is a stronger condition than being a normal subgroup which only requires invariance under inner automorphisms. If $G \cong C_2 \times C_2$, then G has no characteristic subgroups, as there is an automorphism swapping the two C_2 factors, so this is a strictly stronger condition than just being normal.

Proposition 1.1.4. If $M, N \leq G$ and $M \text{ char } N$, $N \trianglelefteq G$, then $M \trianglelefteq G$.

Proof. If M is just normal in G then we don't know it to be invariant under all conjugations, just those from N , but now G is closed under all automorphisms of N , this includes conjugation from elements of G , so M is normal. $\square$

Proposition 1.1.5. The commutator $G' := [G, G]$ is a characteristic subgroup of G , and G/G' is abelian. For any normal subgroup $N \trianglelefteq G$, G/N is abelian iff $G' \leq N$.

Proof. $\square$

Proposition 1.1.6. If $A, B \trianglelefteq G$, $A \cap B = \{1\}$, and $G = AB$, then $G \cong A \times B$.

Theorem 1.1.7. If G is a finitely generated abelian group, then there exists a positive integer $m \geq 0$ and prime powers $p_1^{\alpha_1}, \dots, p_n^{\alpha_n}$ such that:

$$G \cong \mathbb{Z}^m \times C_{p_1^{\alpha_1}} \times \cdots \times C_{p_n^{\alpha_n}}$$

said m and the $p_i^{\alpha_i}$ are uniquely determined (up to reordering of the prime powers) by G .

1.2 Actions

Proposition 1.2.1. If $G \curvearrowright \Omega$, then for any $\alpha \in \Omega$, the size of the orbit α^G equals $|G : G_\alpha| = |G|/|G_\alpha|$ where $G_\alpha \leq G$ is the stabiliser of α in G .

Corollary 1.2.2 (Cauchy's theorem). If G is a finite group whose order is divisible by a prime p , then G contains an element of order p

2 Composition Series

3 Finite simple groups

4 Sylow's Theorems

5 Extensions